

Description of Additional Supplementary Files

File Name: Supplementary Data 1

Description: The table shows the details of the 11 reward schedules tested in the competition. Schedule IDs (first column) correspond to the schedule IDs shown in Fig S1. We asked participants to report whether they used qualitative or quantitative approaches in designing the reward schedules, making them choice architects or choice engineers as indicated in the third column. In addition, participants provided a short explanation of the method they used to create the schedule. An additional reward schedule (Model ID: 0) was designed by an iterative process to estimate the maximally attainable bias (see Results). The table includes four QL models whose parameters were fitted from two datasets. The dataset named ‘online’ was published with the announcement of the competition, and was collected in an online experiment identical to the one used in the competition. The dataset named ‘lab’ was collected in a lab setting in an experiment that is conceptually similar to the experiment used in the competition. Each of these datasets includes many participants whose parameters were fitted to a QL model independently per participant. In ‘homogeneous’ QL models, the collection of individual sets of parameters was averaged across participants. In ‘heterogeneous’ QL models, the individual sets of parameters were used as an ensemble, with the fitted parameters of an individual participant randomly drawn for a full reward schedule. P-values, d-values (effect size), and 95% confidence intervals compare the bias distribution of the winning schedule (Schedule 1) with the rest of the submitted schedules (see Methods).

File Name: Supplementary Data 2

Description: Data were collected from 695 participants, and they were tested on six dynamic schedules. We excluded from the analysis 19 participants (2.7%) that chose one of the alternatives less than 5 times and are therefore likely to have ignored the reward schedule. Adding them to the analysis does not change the ranking of the schedules and their choices are posted together with those of the included participants. We did not collect any demographic information about the participants.

The number of possible dynamic schedules is far larger than that of static schedules (about 101500 larger). Therefore, the ability to bias choices with a dynamic schedule is far larger than this ability with a static schedule. Indeed, the dynamic schedules were more effective than the static ones. Regrettably, only choice architects participated in the dynamic competition and therefore, we could not use it to compare choice architects to choice engineers. The winning dynamic schedule was proposed by Oren Amsalem and Haran Shani.